

Blender for 3D Printing

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Blender for 3D Printing

What is Blender? <https://www.blender.org/>

Importing 3D models

Editing 3D models

Sculpting in Blender

Principles of 3D printing

Designing for 3D printing

Slicing for 3D printing

What is Blender?

Blender <https://www.blender.org/>

Mission

Get the world's best 3D CG technology in the hands of artists as free/open source software.

Mesh-based

Best for artistic/sculptural models

Plug-ins allow many new operations

Artist

FreeCAD <https://www.freecad.org/>

*FreeCAD is a powerful and evolving open-source **parametric 3D modeler**, well-suited for many professional and technical tasks.*

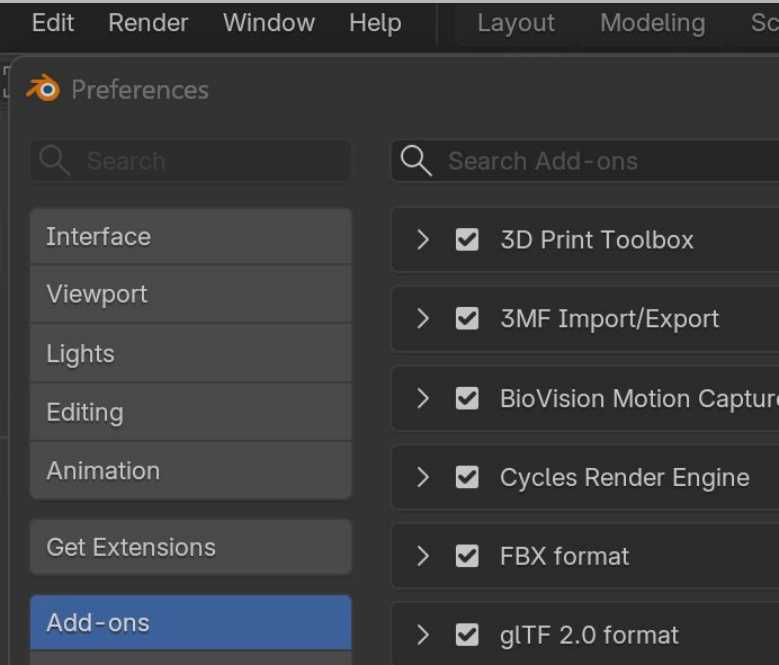
Based on primitives and operations

Great for designing things that need precise measurements. Easy to change components.

Engineer

Artist-Engineer = Maker

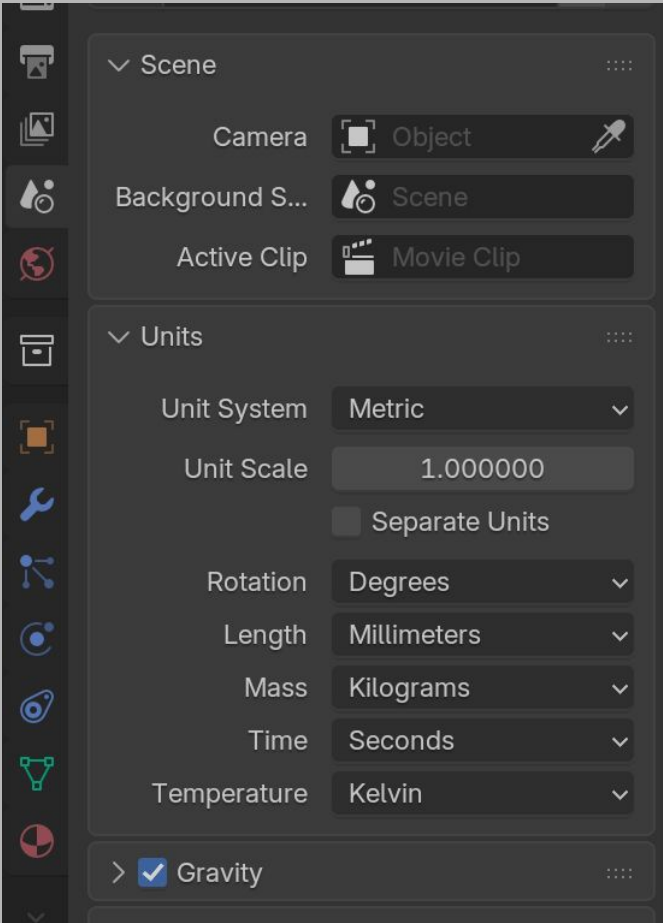
Blender setup



Extensions

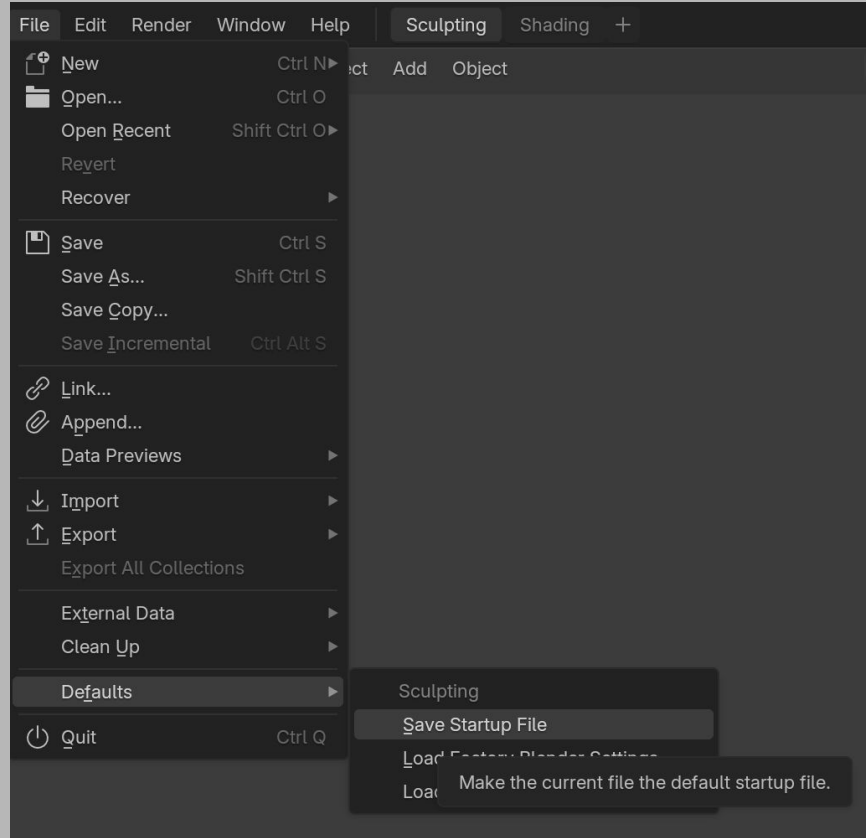
3D Print Toolbox

3MF Import/Export

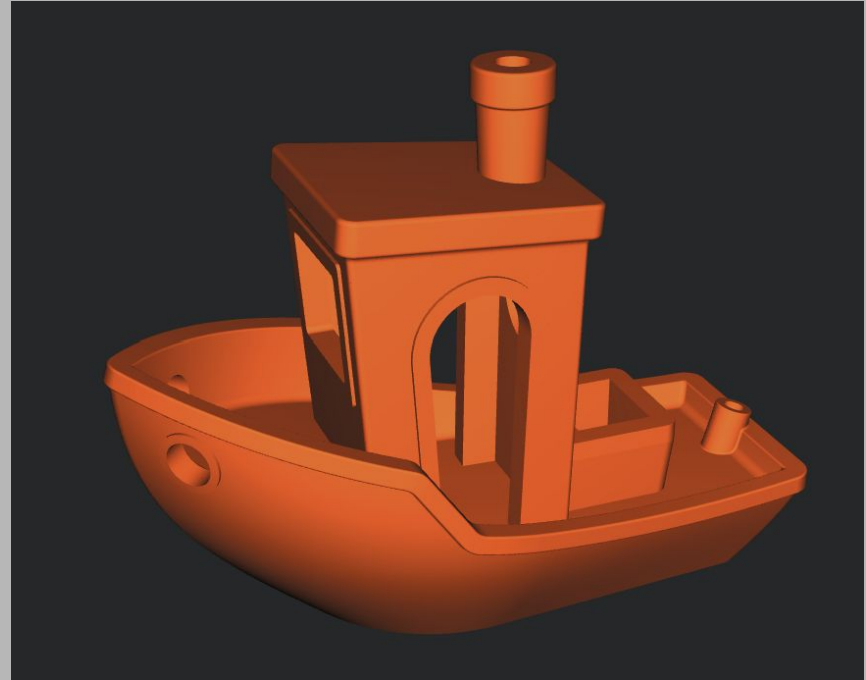
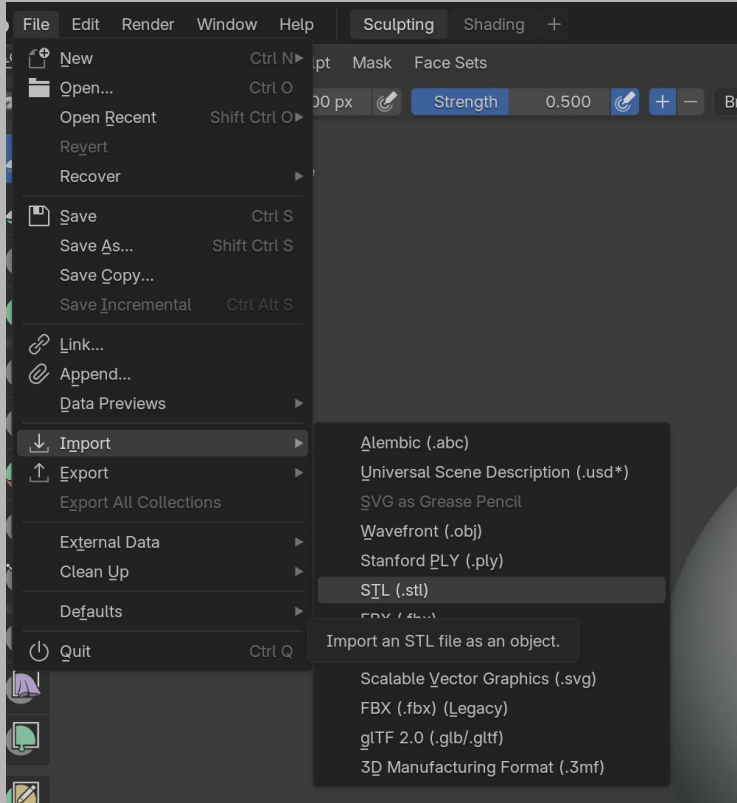


Units -> mm

Save your default startup file for sculpting 3d



Importing 3D models

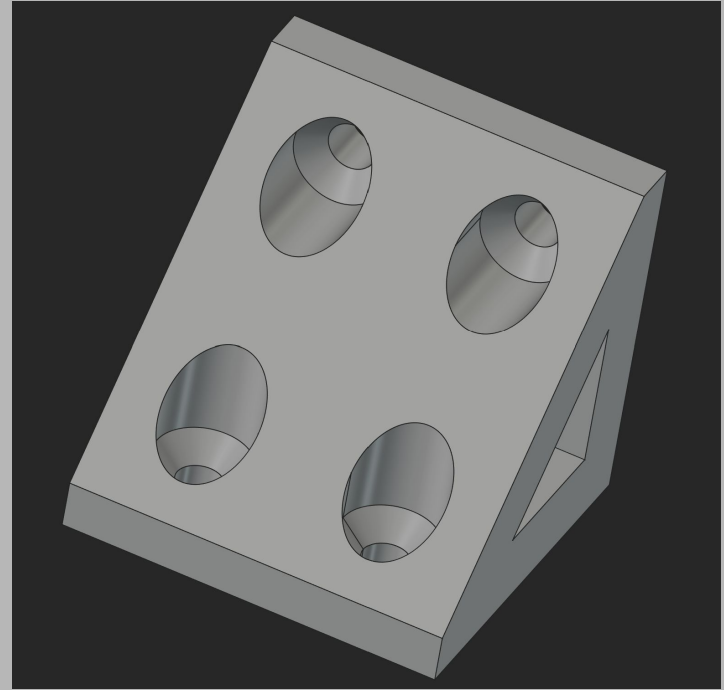
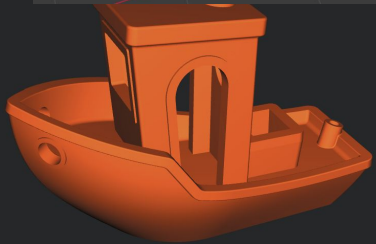
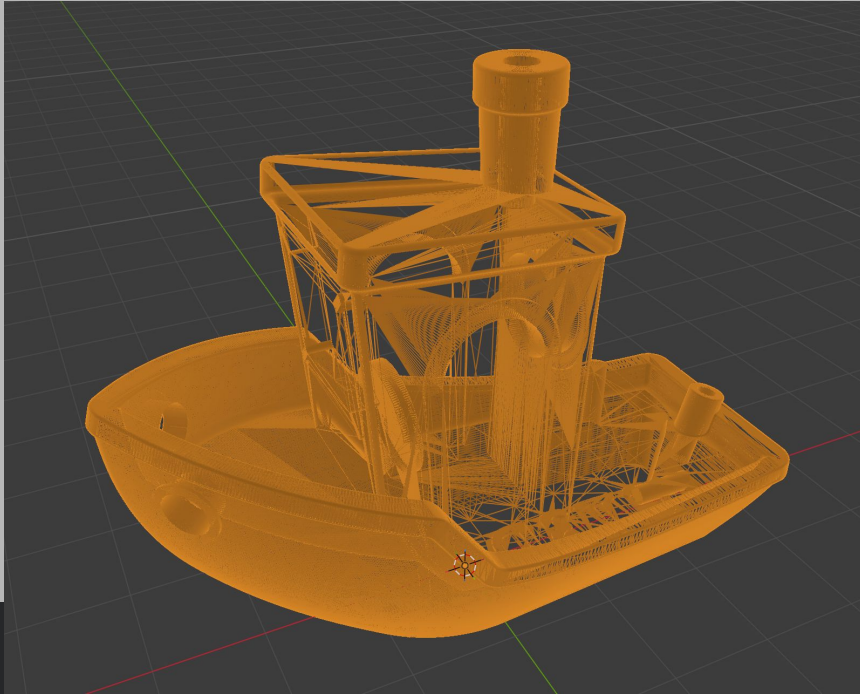


3dbenchy.stl

<https://www.printables.com/model/3161-3d-benchy>

Exercise:
Import an STL file into Blender
from the Internet

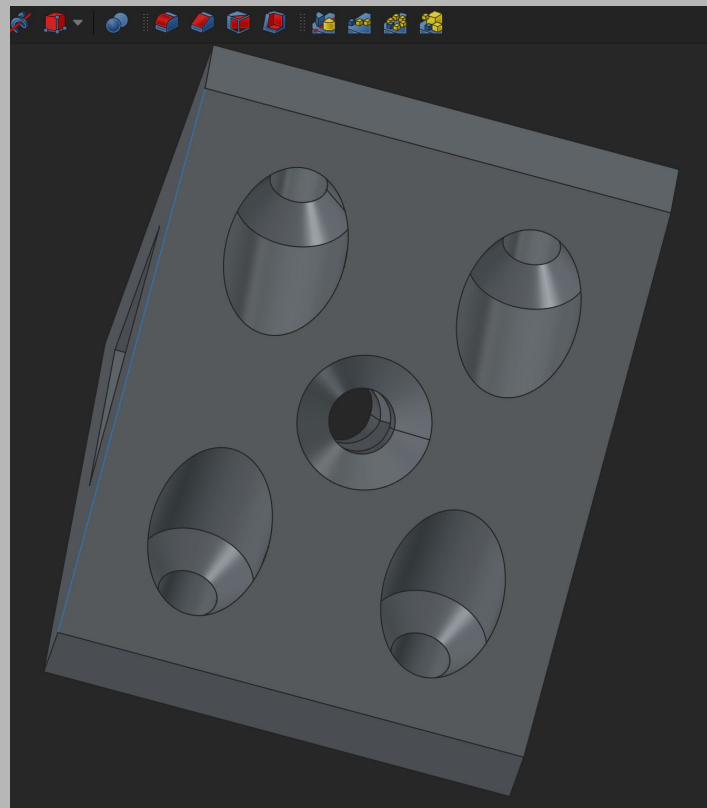
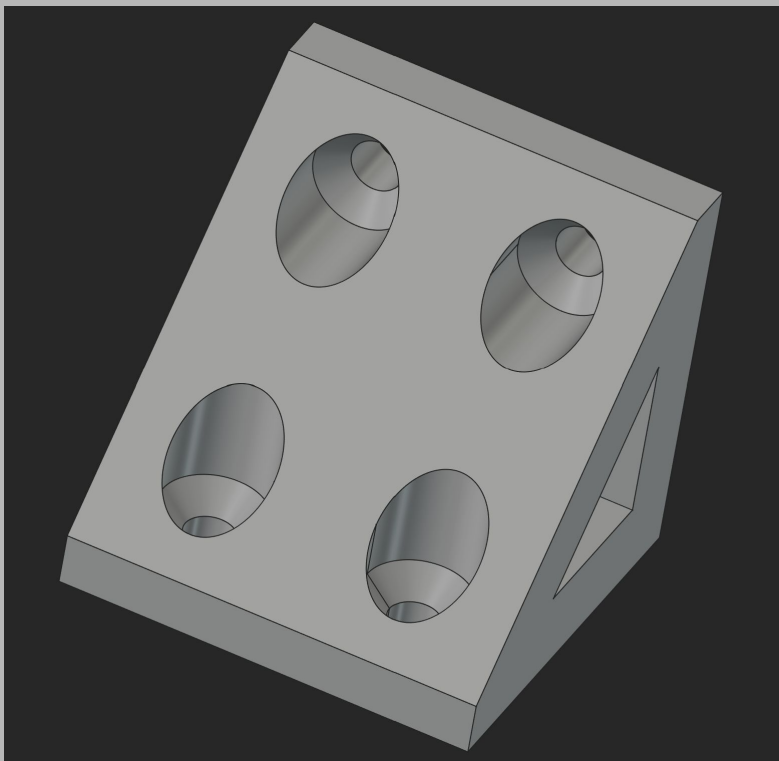
STL vs STEP



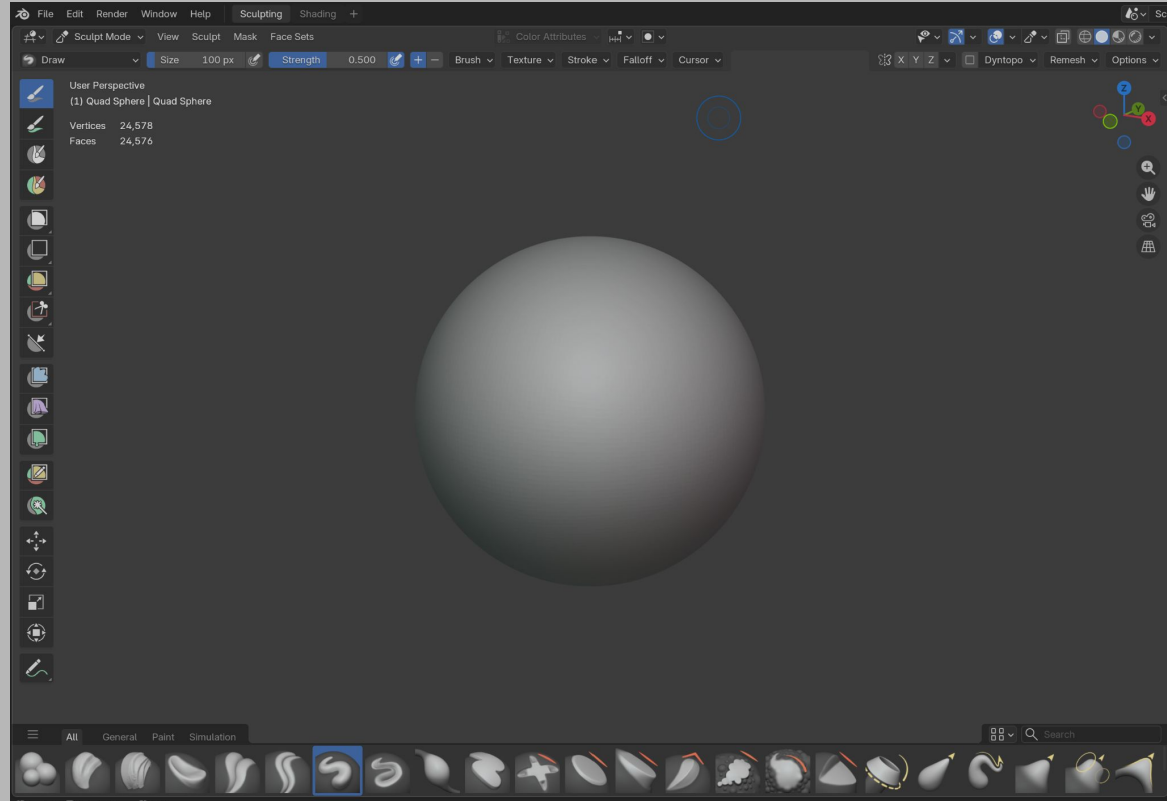
cornerbracket.stp

<https://www.printables.com/model/1803757-stro-nq-corner-bracket/files>

Why STEP?



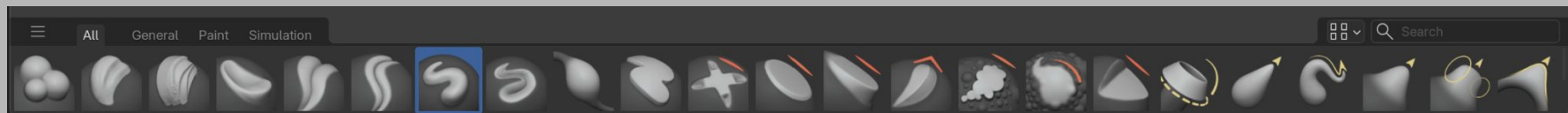
Editing 3D models in Blender



The Sculpting Tools

Ignore

Play with these



Exercise:
make a sculpture from scratch by
sculpting basic shapes

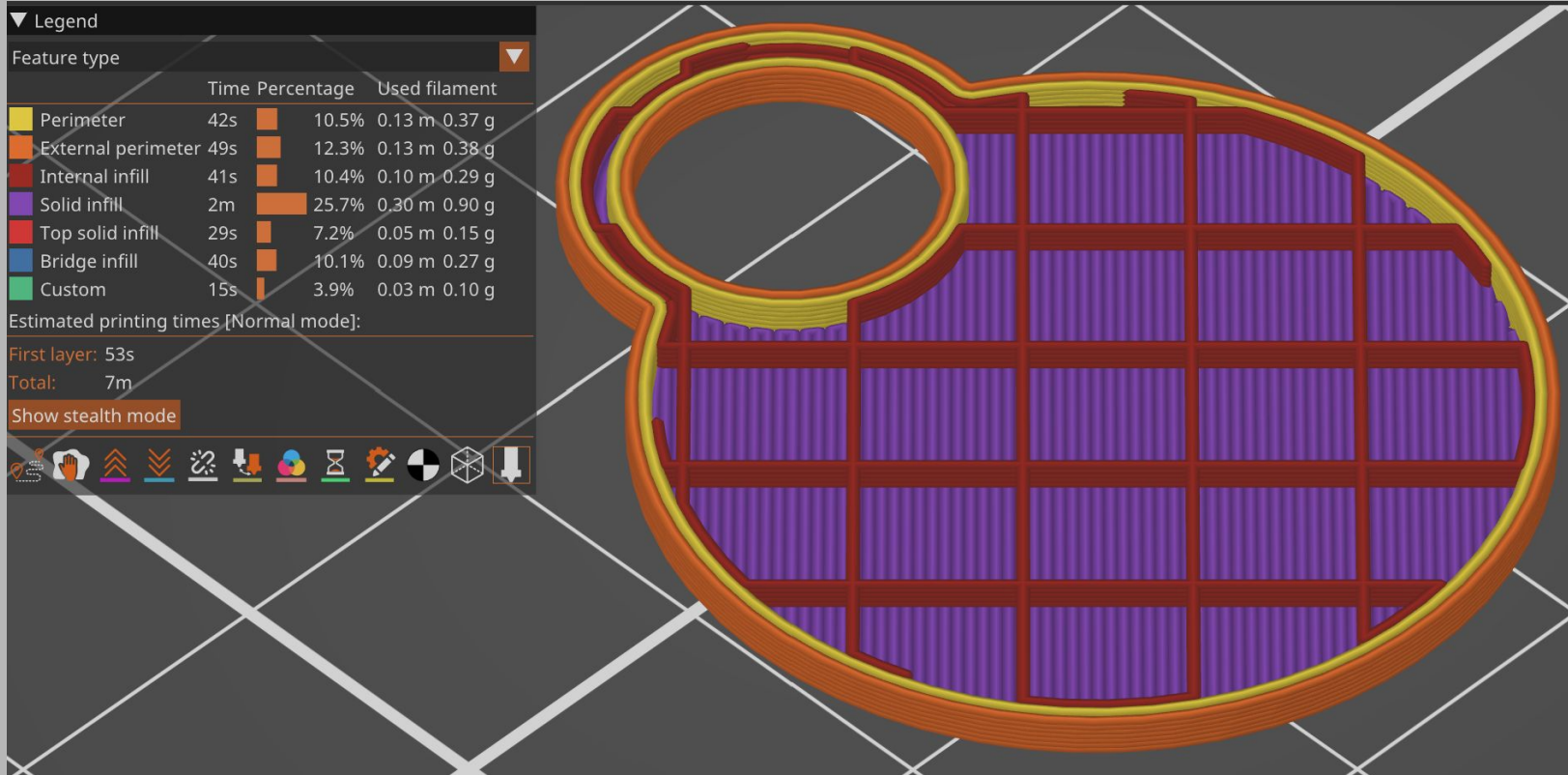
Principles of 3d printing

What is additive manufacturing?

Build up layer by layer

Filament vs resin

Bottom layers



Middle layers

▼ Legend

Feature type

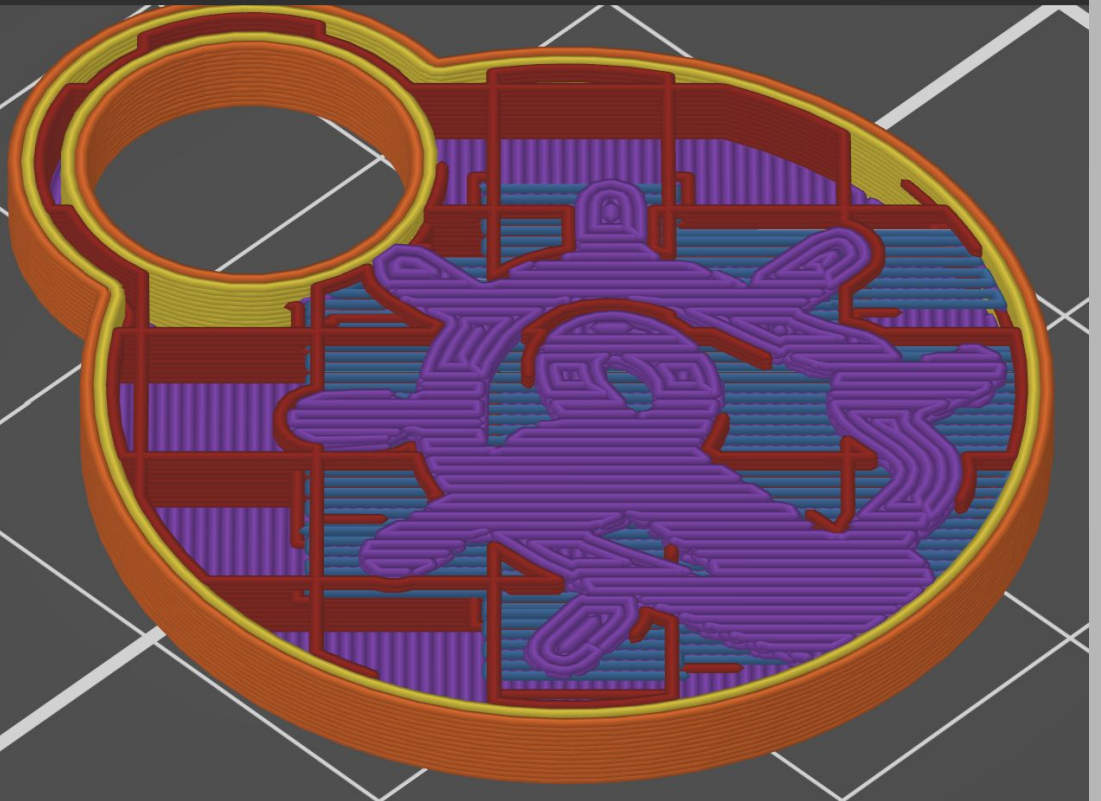
	Time	Percentage	Used filament
Perimeter	42s	10.5%	0.13 m 0.37 g
External perimeter	49s	12.3%	0.13 m 0.38 g
Internal infill	41s	10.4%	0.10 m 0.29 g
Solid infill	2m	25.7%	0.30 m 0.90 g
Top solid infill	29s	7.2%	0.05 m 0.15 g
Bridge infill	40s	10.1%	0.09 m 0.27 g
Custom	15s	3.9%	0.03 m 0.10 g

Estimated printing times [Normal mode]:

First layer: 53s

Total: 7m

Show stealth mode



Top layer

▼ Legend

Feature type

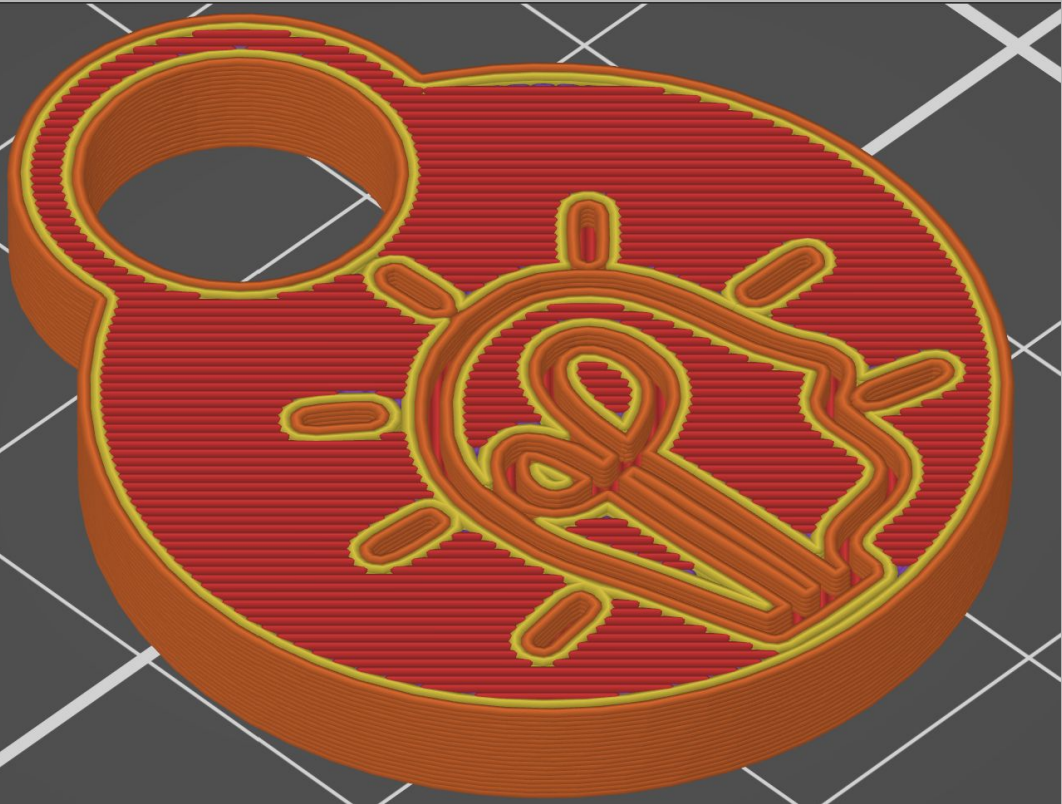
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Filament vs Resin

Layer by layer

Melts the filament in a hot end and lays down the layer

Advantages

Stronger (usually)

No mess (usually)

Faster

Disadvantages

Lower resolution (but good enough for most things)

Layer by layer

UV light used to cure liquid resin so it sticks to previous layer

Advantages

Higher resolution

Useful for smaller models

Disadvantages

Toxic/flammable liquids

Designing for 3D printing

Sculpture vs Part?

Supports

Overhangs

Detail level

Break

Slicing for 3D printing using Prusaslicer

support material

infill types

top/bottom pattern

ironing

print bed texture

adding/subtracting parts

adding text

add support

cutting for smaller print
volumes

Vase mode

Exercise:

Download an STL model of your choice.
Personalize using only the slicer.

Review and Questions

Thanks!

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<https://monumentalimpact.org/>